LAB WORKSHEET

PC Build Lab

Assemble a PC safely and in the right order, diagnose POST faults and choose compatible parts.

Build order

1. Wrist strap on, PSU off.
2. I/O shield, standoffs, motherboard.
3. CPU (triangles matched), close the lever.
4. Thermal paste, then the cooler (fan to CPU_FAN).
5. RAM in A2 + B2 (dual channel); M.2 NVMe SSD.
6. GPU in the top x16 slot.
7. PSU, drives, fans.
8. Cables: 24-pin ATX, 8-pin EPS, PCIe 6+2, SATA power + data, front panel.
9. Power on → POST → UEFI → install the OS.

Debug LEDs

- **CPU**: no EPS power, CPU not seated, no cooler.
- **DRAM**: no RAM / not seated.
- **VGA**: "F" CPU with no GPU, or no PCIe power.
- **BOOT**: no drive detected.

Compatibility

Socket and RAM type must match the board; the case must fit the board and GPU; PSU ≥ (CPU + GPU + 50 W) × 1.3.

Instructor: Aaron Cocker-Swanick